Pseudo-code for Transient Brokerage Simulation

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, matching, or client-demand acquisition rules. Dark-red notation and algorithm lines are specific to the optional client-demand acquisition model.

For any matrix Z, $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L. For a vector z, z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
Core		
N	scalar	Number of agents, with agent set $[N] = \{1, \ldots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t.
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \ldots; m^T]$	$\mathbb{R}^{T\times m}$	Measure table; row $m^t = \mathcal{M}_{t\cdot} \in \mathbb{R}^m$.
Parameters		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction/confidence EWMA weight.
E_{init}	scalar	Initial training steps.
η_{lr}	scalar	MLP learning rate.
Agent types, networks, and output		
$G^t = (V, \mathcal{E}^t_G)$	graph	Undirected graph, $V = [N] \cup \{N+1\}$; node $N+1$ is the broker.
$X^t = [x^t_1, \ldots, x^t_N]$	$\mathbb{R}^{d\times N}$	Agent type matrix; $x^t_i = X^t_{\cdot i} \in \mathbb{R}^d$.
γ	$[0,1] \to \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q^t_{ij}	scalar	Realized output for match (i, j).
True matching function		
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Regime-gain amplitude.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d\times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d\times d}$	Symmetric regime operator.
Calibration and costs		
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.

Symbol	Dimension/domain	Definition
ϕ	scalar	Standard broker fee.
c_s	scalar	Self-search cost.
Learning and predictions		
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d\times n_i^t}$, $y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\cdot\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (\Xi_b^t, y_b^t, n_b^t)$	$\Xi_b^t \in \mathbb{R}^{2d\times n_b^t}$, $y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ has ordered-pair input $\Xi_{b,\cdot\ell}^t = [x_u; x_v]$ and output $y_{b\ell}^t$.
Θ_i^t	NN parameters	Agent i MLP weights.
Θ_b^t	NN parameters	Broker MLP weights.
$v_b^t(i,j)$	scalar	Symmetric broker prediction for unordered pair $\{i,j\}$.
$w_{j\leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_{i\cdot}^t$	vector	Agent i's partner-indexed empirical means.
$C_{i\cdot}^t$	vector	Agent i's partner-indexed counts.
Period state and market objects		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i.
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records: standard $(i, j, z, 0, q, \hat{q}, o_1, o_2)$; principal $(i, j, z, 1, q, \hat{q}, p, o_1, o_2)$.
Capture readiness		
κ_b^t	scalar	EWMA broker exposure MAE.
$n_{\kappa,b}^t$	integer	Cumulative broker exposure-error count.
CapReady_b^t	binary	Principal-mode readiness gate.

The mutable simulation state at the end of period t is

$$S^t = \Big(X^t, G^t, \gamma, \{\mathcal{H}_i^t, \Theta_i^t, \bar{q}_{i\cdot}^t, C_{i\cdot}^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \mathcal{H}_b^t, \Theta_b^t, \text{Roster}^t, D^t, \text{rep}_b^t, \kappa_b^t, n_{\kappa,b}^t\Big).$$

The matching function is

$$f(x_i, x_j) = \rho\,\frac{x_i^\top c + x_j^\top c}{2} + (1-\rho)\,g(x_i, x_j)\,x_i^\top A x_j, \qquad g(x_i, x_j) = 1 + \delta\,\operatorname{sign}(x_i^\top B x_j).$$

2 Top-Level Simulation

Algorithm 1 Transient Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and measure table $\mathcal{M} \in \mathbb{R}^{T\times m}$.

$S^0 \leftarrow \textsc{Initialize}(\vartheta, \text{seed})$

for $t = 1, \ldots, T$ **do**

3: $(S^t, \mathcal{R}^t) \leftarrow$ PeriodUpdate(S^{t-1}, t, ϑ)
4: $\mathcal{M}_{t\cdot} \leftarrow$ CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$
5: **return** $(S^T, \mathcal{M})$

3 Training

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h,

$$\mathrm{NN}_\Theta(\xi) = w_2^\top \operatorname{ReLU}(W_1\xi + b_1) + b_2, \qquad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h\times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$. Agent networks use $(p_{\text{in}}, h) = (d, h_a)$; the broker network uses $(p_{\text{in}}, h) = (2d, h_b)$. For broker symmetry augmentation, let

$$\mathsf{P}_d = \begin{bmatrix} 0_{d\times d} & I_d \\ I_d & 0_{d\times d} \end{bmatrix}, \qquad \mathsf{P}_d[a; b] = [b; a] \quad (a, b \in \mathbb{R}^d).$$

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
Θ	NN parameters	Generic MLP parameters (W_1, b_1, w_2, b_2).
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
h_a, h_b	integers	Agent and broker hidden widths.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}}\times n}$, $\mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
e	integer	Gradient-step index.
W	integer	Training-window length.
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t.
L_i, L_b	ordered index sets	Training-window observation indices.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker gradient-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d\times\lvert L_i\rvert}$, $\mathbb{R}^{\lvert L_i\rvert}$	Agent i training view of $\mathcal{H}_i^t$.
$\Xi_{b,\text{tr}}, y_{b,\text{tr}}$	$\mathbb{R}^{2d\times 2\lvert L_b\rvert}$, $\mathbb{R}^{2\lvert L_b\rvert}$	Symmetry-augmented broker training view of $\mathcal{H}_b^t$.

Algorithm 2 TrainNN

Require: Network parameters Θ, inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}}\times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E, learning rate η_{lr}.
1: **for** $e = 1, \ldots, E$ **do**
2: $\mathcal{L}(\Theta) \leftarrow n^{-1}\sum_{\ell=1}^{n} \left(\mathrm{NN}_\Theta(\Xi_{\text{tr},\cdot\ell}) - y_{\text{tr},\ell}\right)^2$
3: $\Theta \leftarrow \Theta - \eta_{\text{lr}}\nabla_\Theta\mathcal{L}(\Theta)$
4: **return** Θ

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 3 TrainPredictors

Require: State S, period t, parameters $(E_{\text{init}}, \eta_{\text{lr}})$.

$W \leftarrow 500$
for $i = 1, \ldots, N$ in parallel **do**
    $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
    if $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
        $L_i \leftarrow$ ordered indices of the most recent $\min(W, n_i)$ observations in $\mathcal{H}_i$.
        $\Xi_{i,\mathrm{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\mathrm{tr}} \leftarrow (y_i^t)_{L_i}$.
        $E_i \leftarrow \max\{50, \lceil E_{\mathrm{init}} \Delta_i / n_i \rceil\}$.
        $\Theta_i \leftarrow \textsc{TrainNN}(\Theta_i, \Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}, E_i, \eta_{\mathrm{lr}})$; $\Delta_i \leftarrow 0$.
$n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
if $n_b > 0$ and $\Delta_b > 0$ **then**
    $L_b \leftarrow$ ordered indices of the most recent $\min(W, n_b)$ observations in $\mathcal{H}_b$.
    $\Xi_{b,\mathrm{tr}} \leftarrow \left[(\Xi_b^t)_{\cdot L_b}, \mathsf{P}_d(\Xi_b^t)_{\cdot L_b}\right]$.
    $y_{b,\mathrm{tr}} \leftarrow \left((y_b^t)_{L_b}, (y_b^t)_{L_b}\right)$.
    $E_b \leftarrow \max\{50, \lceil E_{\mathrm{init}} \Delta_b / n_b \rceil\}$.
    $\Theta_b \leftarrow \textsc{TrainNN}(\Theta_b, \Xi_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}, E_b, \eta_{\mathrm{lr}})$; $\Delta_b \leftarrow 0$.

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\mathrm{cal}} = \frac{1}{10{,}000} \sum_{\ell=1}^{10{,}000} \{Q + f(x_{i_\ell}, x_{j_\ell})\}, \qquad r = 0.60\bar{q}_{\mathrm{cal}}, \qquad \phi = c_s = \lambda_c(\bar{q}_{\mathrm{cal}} - r).$$

The capture-readiness gate is

$$\mathrm{CapReady}_b^t = \begin{cases} 1, & n_{\kappa,b}^t \geq n_{\min} \quad \text{and} \quad \kappa_b^t/(\bar{q}_{\mathrm{cal}} - r) \leq \kappa_{\max}, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathrm{CapReady}_b^0 = 0.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s.
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
$n_{\min}, \kappa_{\max}$	scalars	Principal-mode confidence gate.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ.
ψ_ℓ	scalar	Phase for active curve coordinate ℓ.
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A.
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization regime matrix.
$\Xi_{b,\mathrm{seed}}, y_{b,\mathrm{seed}}$	$\mathbb{R}^{2d \times 2n_b^0}$, $\mathbb{R}^{2n_b^0}$	Symmetry-augmented broker seed training batch.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i.
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 4 InitializeAgent

Require: State S, agent slot i, period counter t, parameters ϑ.
Ensure: Agent slot i is initialized in S.

Draw $u_i \sim \text{Unif}[0,1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
$x_i^t \leftarrow (\gamma(u_i) + \epsilon_i)/\|\gamma(u_i) + \epsilon_i\|_2$.
Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_{i\cdot}^t$, $C_{i\cdot}^t$, χ_i^t, and M_i^t; draw fresh agent MLP weights Θ_i^t with output bias Q.
if $t = 0$ **then**
  $s_i^t \leftarrow (0,0)$.
else
  Remove i from Roster^t, D^t, and M_j^t for all j; delete all incident edges in G^t; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
  Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

  Add edges (i,j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.

$$s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$$

  $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.

Algorithm 5 Initialize

Require: Parameters ϑ and seed.
Ensure: Initial state S^0.

Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1,\ldots,5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \ldots, d_\gamma$, defining γ.
Allocate empty agent and broker containers in S^0 and store γ.
for $i = 1, \ldots, N$ **do**
  $\textsc{InitializeAgent}(S^0, i, 0, \vartheta)$.
Draw ideal agent type c as a noisy curve point.
Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \operatorname{tr}(A_0^\top A_0)$.
$\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\operatorname{tr}(\Sigma_x B_0 \Sigma_x A)}{\operatorname{tr}(\Sigma_x A \Sigma_x A)} A, \qquad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.

Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire}; add isolated broker node $N+1$.
Initialize $\mathcal{H}_b^0$, Θ_b^0, and rep_b^0.
Initialize capture gate: $\kappa_b^0 \leftarrow 0$, $n_{\kappa,b}^0 \leftarrow 0$, and $\text{CapReady}_b^0 \leftarrow 0$.
Sample $\text{Roster}^0 \subset [N]$ uniformly with $|\text{Roster}^0| = \lceil 0.2N \rceil$; add broker edges $(i, N+1)$ for all roster members.
for each unordered non-broker edge $\{i,j\} \in \mathcal{E}_G^0$ **do**
    Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.
Let $\mathcal{B}_{\text{seed}} \leftarrow \{\{i,j\} \in \mathcal{E}_G^0 : i,j \in \text{Roster}^0\}$; shuffle and keep at most 100 edges.
for each $\{i,j\} \in \mathcal{B}_{\text{seed}}$ **do**
    Append column $[x_i^0; x_j^0]$ to Ξ_b^0; append the existing edge outcome q_{ij} to y_b^0; increment n_b^0.
if $n_b^0 > 0$ **then**
    Set rep_b^0 to mean broker seed output.
else
    Set $\text{rep}_b^0 \leftarrow 0$.
Set $s_{i,\text{self}}^0$ to mean own seed output and $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$.
for each i with $n_i^0 > 0$ **do**
    $\Theta_i^0 \leftarrow \textsc{TrainNN}(\Theta_i^0, \Xi_i^0, y_i^0, E_{\text{init}}, \eta_{\text{lr}})$.
$\Xi_{b,\text{seed}} \leftarrow [\Xi_b^0, \mathsf{P}_d \Xi_b^0]$, $y_{b,\text{seed}} \leftarrow (y_b^0, y_b^0)$.
$\Theta_b^0 \leftarrow \textsc{TrainNN}(\Theta_b^0, \Xi_{b,\text{seed}}, y_{b,\text{seed}}, E_{\text{init}}, \eta_{\text{lr}})$.
return S^0.

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 6 ChooseChannel

Require: Agent i state (s_i, χ_i), broker reputation rep_b, RNG.
$U_{\text{self}} \leftarrow s_{i,\text{self}}$.
$U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1-\chi_i)\text{rep}_b$.
if $U_{\text{self}} > U_{\text{broker}}$ **then**
    return self.
if $U_{\text{broker}} > U_{\text{self}}$ **then**
    return broker.
return self w.p. 1/2, else broker ▷ $U_{\text{self}} = U_{\text{broker}}$.

6 One Period

Local variables for PeriodUpdate		
Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 7 PeriodUpdate

Require: Previous state S^{t-1}, period t, parameters ϑ.
Ensure: Updated state S^t and accepted records $\mathcal{R}^t$.

Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i, and set $D^t \leftarrow \emptyset$.
Refresh roster: remove each roster member with probability p_{roster}, then replenish uniformly to size $\lceil 0.2N \rceil$.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
$\mathcal{D}^t \leftarrow \emptyset$.
for $i = 1, \ldots, N$ **do**
    Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
    if $d_i^t > 0$ **then**
        $z_i^t \leftarrow \textsc{ChooseChannel}(i, \text{rep}_b^{t-1})$.
        Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
        if $z_i^t = \text{broker}$ **then**
            Add i to D^t.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
$\textsc{TrainPredictors}(S, t, \vartheta)$.
$\mathcal{R}^t \leftarrow \textsc{OfferMarket}(S, \mathcal{D}^t, \vartheta)$.
$\textsc{UpdateSatisfaction}(S, \mathcal{D}^t, \mathcal{R}^t)$.
if $D^t \neq \emptyset$ **then**
    $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i,\text{broker}}^t$.
else
    $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
$\textsc{UpdateBrokerConfidence}(S, \mathcal{R}^t)$.
if $t \bmod \Delta_{\text{net}} = 0$ **then**
    Recompute broker betweenness, Burt constraint, and effective size on G^t.
for $i = 1, \ldots, N$ **do**
    if agent i exits with probability η_{exit} **then**
        $\textsc{InitializeAgent}(S^t, i, t, \vartheta)$.
return $(S^t, \mathcal{R}^t)$.

7 Offer Market

Local variables for OfferMarket		
Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \to \mathbb{R}$	Agent predictor.

Local variables for OfferMarket (continued)		
Symbol	Dimension/domain	Definition
$\hat{q}_b^t$	$\mathbb{R}^{2d} \to \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i.
U^t	subset of $[N]$	Period stranger pool.
$n_{\text{strangers}}$	integer	Period stranger-pool size.
$v_{i\to j}^{\text{self},t}$	scalar	Self-offer score.
Ω_i	integer	Residual offer quota for demander i.
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \to j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t.
$\mathcal{Z}^t$	subset of $[N]$	Acquired clients removed from standard offers.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.
$\mathcal{I}_b$	list	Broker pairs sorted by v_b^t.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and o.from.
$z^\star$	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i, j) = \frac{1}{2}\left[\hat{q}_b^t([x_i^t; x_j^t]) + \hat{q}_b^t([x_j^t; x_i^t])\right].$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined only for observed graph neighbors and period strangers:

$$v_{i\to j}^{\text{self},t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i),\ C_{ij}^t > 0, \\ \hat{q}_i^t(x_j^t), & j \in U^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means dominate predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j\leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t,\ C_{ji}^t > 0, \\ \hat{q}_j^t(x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i. Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 8 AddOffer

Require: Offer book O^t, unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v).

if O_{ij}^t is defined **then**

return 0.
$O^t_{ij} \leftarrow (i, j, z, v)$.
if O^t_{ji} is undefined **then**
$\mathcal{P}^t_O \leftarrow \mathcal{P}^t_O \cup \{\{i, j\}\}$.
return 1.

Algorithm 9 OfferMarket

Require: State S, demand records $\mathcal{D}^t$, parameters ϑ.
Ensure: Accepted records $\mathcal{R}^t$.
For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
$O^t \leftarrow \emptyset$, $\mathcal{P}^t_O \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
Draw one period-level stranger pool $U^t \subset [N]$ uniformly without replacement, $|U^t| = \min(n_{\text{strangers}}, N)$.
$\mathcal{Z}^t \leftarrow \emptyset$.
if principal mode is enabled **then**
$\mathcal{Z}^t \leftarrow \textsc{AcquireClientDemand}(S, \mathcal{D}^t, \Omega, \mathcal{R}^t, \vartheta)$.
for each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
$\mathcal{U}^{\text{self}}_i \leftarrow \{j \in \mathcal{N}^t_G(i) : C^t_{ij} > 0\} \cup (U^t \setminus \mathcal{N}^t_G(i))$.
Remove from $\mathcal{U}^{\text{self}}_i$ the elements of $\{i\} \cup M^t_i \cup \mathcal{Z}^t$.
$\mathcal{I}^{\text{self}}_i \leftarrow \text{argsort}_{j \in \mathcal{U}^{\text{self}}_i :\ v^{\text{self},t}_{i \to j} > r}(-v^{\text{self},t}_{i \to j})$.
for the first $\min(\Omega_i, |\mathcal{I}^{\text{self}}_i|)$ agents j in $\mathcal{I}^{\text{self}}_i$ **do**
$\textsc{AddOffer}(O^t, \mathcal{P}^t_O, i, j, \text{self}, v^{\text{self},t}_{i \to j})$.
$\mathcal{A}^t_b \leftarrow (\text{Roster}^t \cup D^t) \setminus \mathcal{Z}^t$.
$D^t_+ \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t,\ i \notin \mathcal{Z}^t,\ \Omega_i > 0\}$.
$\mathcal{P}^t_b \leftarrow \{\{i, j\} : i \neq j,\ (i \in D^t_+, j \in \mathcal{A}^t_b) \text{ or } (j \in D^t_+, i \in \mathcal{A}^t_b)\}$.
$\mathcal{I}_b \leftarrow \text{argsort}_{\{i,j\} \in \mathcal{P}^t_b : v^t_b(i,j) > r}(-v^t_b(i, j))$.
for each $\{i, j\} \in \mathcal{I}_b$ **do**
for each ordered pair $(\alpha, \beta) \in \{(i, j), (j, i)\}$ **do**
if $\alpha \in D^t_+$, $\beta \in \mathcal{A}^t_b$, and $\Omega_\alpha > 0$ **then**
$\iota \leftarrow \textsc{AddOffer}(O^t, \mathcal{P}^t_O, \alpha, \beta, \text{broker}, v^t_b(i, j))$.
$\Omega_\alpha \leftarrow \Omega_\alpha - \iota$.
for each $\{i, j\} \in \mathcal{P}^t_O$ **do**
$o_{ij} \leftarrow O^t_{ij}$ if defined, else $\emptyset$; $o_{ji} \leftarrow O^t_{ji}$ if defined, else $\emptyset$.
if $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
$\textsc{FinalizeStandard}(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
else if $o_{ij} \neq \emptyset$ and $w^t_{j \leftarrow i} > r$ **then**
$\textsc{FinalizeStandard}(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
else if $o_{ji} \neq \emptyset$ and $w^t_{i \leftarrow j} > r$ **then**
$\textsc{FinalizeStandard}(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
return $\mathcal{R}^t$.

Algorithm 10 FinalizeStandard

Require: State S, accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o'.
if $j \in M^t_i$ **then**
return .
Draw $q^t_{ij} = Q + f(x^t_i, x^t_j) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2_\varepsilon)$.

Append columns x_j^t to Ξ_i and x_i^t to Ξ_j; append q_{ij}^t to y_i and y_j; increment n_i and n_j.
Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
Add edge (i, j) to G^t; add j to M_i^t and i to M_j^t.
if $z = \text{broker}$ or ($o' \neq \emptyset$ and $o'.z = \text{broker}$) **then**
    Append column $[x_i^t; x_j^t]$ to Ξ_b; append q_{ij}^t to y_b; increment n_b.
    $z^\star \leftarrow \text{broker}$.
else
    $z^\star \leftarrow \text{self}$.
Append accepted record $(i, j, z^\star, 0, q_{ij}^t, v, o, o')$ to $\mathcal{R}^t$.

8 Optional Client-Demand Acquisition

Local variables for AcquireClientDemand

Symbol	Dimension/domain	Definition
p_i	scalar	Acquisition price per acquired position.
$\mathcal{J}_i$	subset of $[N]$	Planned recipient set.
Γ_i	scalar	Expected net acquisition gain.
$\mathcal{I}_{\text{acq}}$	list	Feasible client acquisitions sorted by Γ_i.

Algorithm 11 AcquireClientDemand

Require: State S, demand records $\mathcal{D}^t$, residual quotas Ω, accepted records $\mathcal{R}^t$.
Ensure: Acquired clients $\mathcal{Z}^t$; principal records appended to $\mathcal{R}^t$.
if $\text{CapReady}_b^t = 0$ **then**
    return $\emptyset$.
$D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$, $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
Score $\mathcal{P}_b^t = \{\{i, j\} : i \in D_+^t,\ j \in \mathcal{A}_b^t,\ i \neq j\}$ by $v_b^t(i, j)$.
for $i \in D_+^t$ **do**
    $p_i \leftarrow (n_i^t)^{-1} \sum_{\ell=1}^{n_i^t} y_{i\ell}^t$ if $n_i^t > 0$, else $p_i \leftarrow \bar{q}_{\text{cal}}$.
    $\mathcal{J}_i \leftarrow$ top Ω_i distinct $j \in \mathcal{A}_b^t \setminus \{i\}$ with $v_b^t(i, j) > r$.
    $\Gamma_i \leftarrow \sum_{j \in \mathcal{J}_i} v_b^t(i, j) - \Omega_i p_i - \Omega_i \phi$.
$\mathcal{I}_{\text{acq}} \leftarrow \operatorname{argsort}_{i:|\mathcal{J}_i| = \Omega_i,\ p_i \geq r,\ \Gamma_i > 0}(-\Gamma_i)$; $\mathcal{Z}^t \leftarrow \emptyset$.
for $i \in \mathcal{I}_{\text{acq}}$ **do**
    Recompute $\mathcal{J}_i$ from the sorted pair list, excluding current matches and $\mathcal{Z}^t$.
    if $|\mathcal{J}_i| < \Omega_i$ **then**
        continue.
    Add i to $\mathcal{Z}^t$; record principal payment $\Omega_i p_i$ to i; set $\Omega_i \leftarrow 0$.
    for $j \in \mathcal{J}_i$ **do**
        Mark (i, j) as current before receiver evaluation.
        if $w_{j \leftarrow i}^t > r$ **then**
            Draw q_{ij}^t; append column $[x_i^t; x_j^t]$ to Ξ_b; append q_{ij}^t to y_b; increment n_b.
            Add principal active matches $i \leftrightarrow j$.
            Append $(i, j, \text{broker}, 1, q_{ij}^t, v_b^t(i, j), p_i, \emptyset, \emptyset)$ to $\mathcal{R}^t$.
        else
            Record rejected principal exposure with realized value 0 and prediction $v_b^t(i, j)$.

return $\mathcal{Z}^t$.

Principal positions update broker history only when accepted. They do not update agent histories, agent-agent edges, or agent partner means.

9 Satisfaction, Confidence, and Measures

Local variables for satisfaction and confidence

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n^{\mathrm{std}}_{i,b}$	integer	Standard broker offer credits for demander i.
$\tilde{q}^t_i$	scalar	Channel-specific satisfaction input.
$P^{\mathrm{acq},t}_i$	scalar	Principal payment to acquired client i, zero absent acquisition.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.
$\mathcal{E}^t_{\mathrm{err},b}$	set	Broker exposure-error observations.
e^t_b	scalar	Period mean absolute exposure error.

Algorithm 12 UpdateSatisfaction

Require: State S, demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

```
for each (i, z_i, d_i) ∈ D^t do
    Y_i ← Σ_{ζ∈R^t} Σ_{o∈{ζ.o_1, ζ.o_2}} 1{o ≠ ∅, o.from = i, ζ.π = 0} ζ.q + P_i^{acq,t}.
    n_{i,b}^{std} ← Σ_{ζ∈R^t} Σ_{o∈{ζ.o_1, ζ.o_2}} 1{o ≠ ∅, o.from = i, o.z = broker, ζ.π = 0}.
    if z_i = self then
        q̃_i^t ← Y_i/d_i − c_s.
    else
        q̃_i^t ← (Y_i − φ n_{i,b}^{std})/d_i; χ_i^t ← 1.
    s_{i,z_i}^t ← (1 − ω) s_{i,z_i}^{t−1} + ω q̃_i^t.
```

- **for each** $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
  - $Y_i \leftarrow \sum_{\zeta\in\mathcal{R}^t} \sum_{o\in\{\zeta.o_1,\zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\mathrm{from} = i,\ \zeta.\pi = 0\}\, \zeta.q + P^{\mathrm{acq},t}_i$.
  - $n^{\mathrm{std}}_{i,b} \leftarrow \sum_{\zeta\in\mathcal{R}^t} \sum_{o\in\{\zeta.o_1,\zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\mathrm{from} = i,\ o.z = \mathrm{broker},\ \zeta.\pi = 0\}$.
  - **if** $z_i = \mathrm{self}$ **then**
    - $\tilde{q}^t_i \leftarrow Y_i/d_i - c_s$.
  - **else**
    - $\tilde{q}^t_i \leftarrow (Y_i - \phi n^{\mathrm{std}}_{i,b})/d_i$; $\chi^t_i \leftarrow 1$.
  - $s^t_{i,z_i} \leftarrow (1-\omega) s^{t-1}_{i,z_i} + \omega \tilde{q}^t_i$.

Algorithm 13 UpdateBrokerConfidence

Require: State S, accepted records $\mathcal{R}^t$.

- $\mathcal{E}^t_{\mathrm{err},b} \leftarrow \emptyset$.
- **for** each accepted record $\zeta \in \mathcal{R}^t$ **do**
  - **for** each standard offer credit $o \in \{\zeta.o_1, \zeta.o_2\}$ **do**
    - **if** $o \neq \emptyset$ and $o.z = \mathrm{broker}$ **then**
      - Add $(o.v, \zeta.q)$ to $\mathcal{E}^t_{\mathrm{err},b}$.
  - **if** $\zeta.\pi = 1$ **then**
    - Add $(\zeta.\hat{q}, \zeta.q)$ to $\mathcal{E}^t_{\mathrm{err},b}$.
- **if** $|\mathcal{E}^t_{\mathrm{err},b}| > 0$ **then**
  - $e^t_b \leftarrow |\mathcal{E}^t_{\mathrm{err},b}|^{-1} \sum_{(\hat{q},q)\in\mathcal{E}^t_{\mathrm{err},b}} |\hat{q} - q|$.
  - $\kappa^t_b \leftarrow e^t_b$ if $n^{t-1}_{\kappa,b} = 0$, else $(1-\omega)\kappa^{t-1}_b + \omega e^t_b$.
  - $n^t_{\kappa,b} \leftarrow n^{t-1}_{\kappa,b} + |\mathcal{E}^t_{\mathrm{err},b}|$.

CollectMeasures. $\textsc{CollectMeasures}(S^t, \mathcal{R}^t, \vartheta)$ is a pure measure-collection function. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, degree summaries, cached broker network measures, and principal-mode diagnostics when enabled. The measures themselves do not feed back into S^{t+1}.